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The Aging Patient

Paula Ryo, DO
Family Medicine
pryo@nyit.edu

Office of Academic Affairs

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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- By the end, the student will be able to:
 - Describe the basic principles of aging and how they lead to geriatric syndromes that affect patient overall health and wellbeing.
 - Describe some of the effects of aging on the human body.
 - Obtain a history from a geriatric patient.
 - Complete a geriatric assessment

Geriatrics

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- The branch of medicine that focuses on the care of the elderly
 - Focus is on prevention and care of chronic illnesses
- Geriatric patient ≥ 65 yo
 - Oldest Old ≥ 80 yo

Geriatric population

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- The world's population is aging at a rapid rate
- This demographic transition is accompanied by an epidemiologic transition
 - **Demographic imperative:** maximize not only life span but also “that older adults maintain full function as long as possible” (Bates)

SOURCE: Suzman RM, Haaga JG. World Demography of Aging. In: Kasper D, Fauci A, Hauser S, Longo D, Jameson J, Loscalzo J. eds. Harrison's Principles of Internal Medicine, 19e. New York, NY: McGraw-Hill; 2015.
Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

- This demographic transition is accompanied by an epidemiologic transition
 - Non-communicable chronic diseases are becoming the major causes of death and contributors to the burden of disease and disability.
 - Concomitant change in “*dependency*” — the ratio of adults in the workforce to those typically out of the workforce (infants, children, retired “young old” (those still active but in ways other than paid work), the oldest old, etc.)
 - There is a ripple effect on economic growth, migration, patterns of work and retirement, family structures, pension and health systems, etc.

This epidemiologic change and the size of age groups will have effects downstream that will impact healthcare and societal programs. The rate of natural increase of a population depends on birth and death rates, which are strongly influenced by the population age structure. Births occur primarily to people in the younger-adult age groups. If there are comparatively more young adults than older adults, then even at replacement fertility levels (when each woman has about an average of two children) there will be more births than deaths, which means a population that is able to sustain itself and grow accordingly. A population of young people needs a sufficient number of schools and, later, enough jobs to accommodate them. Countries with a large proportion of older people must develop retirement systems and medical facilities to serve them. Therefore, as a population ages, needs change from childcare and schools to jobs, housing, and medical care and the increase in number of geriatric population **sharing a limited number of resources strains the environment so transitions to supported services are necessary: for example:**

Example: ratio of children to older people will affect the relative demand for pediatricians versus geriatricians & the number of hospital beds needed as well as the types of care providers needed will change and the governmental policies and educational structure (medical schools, residencies, fellowships, etc) will also need to change.

Geriatric population worldwide

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- One in eight people are 60yo or older²
- 12.3 % of people worldwide are 60 yrs old and older -> 617 million people²
- By 2050, it is expected to be 22% of the worldwide population²
- The worldwide population of the oldest old (80 or older) is expected to triple by 2050 – from 126.5 million to 446.6 million
- In the US, the 65 and over population is expected to nearly double by 2050 - from 48 million to 88 million

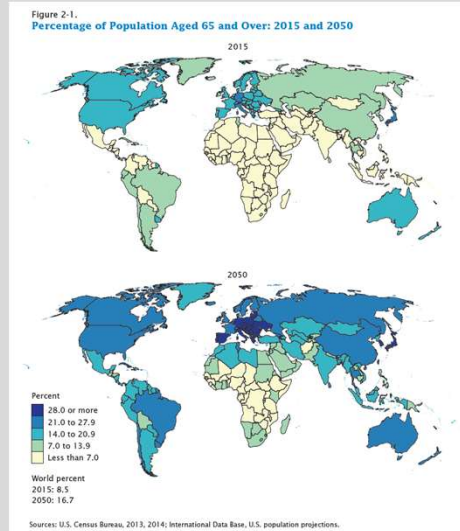
1. He, Wan, Goodkind, Daniel, and Kowal, Paul. An Aging World: 2015 – International population reports. NIH. Issued March 2016.
<https://www.census.gov/content/dam/Census/library/publications/2016/demo/p95-16-1.pdf>

2. United Nations Population Fund, <https://www.unfpa.org/ageing>

Geriatric population - 2015 versus 2050

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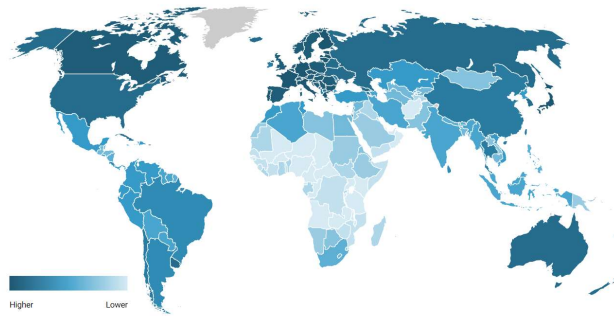
SOURCE: He, Wan, Goodkind, Daniel, and Kowal, Paul. An Aging World: 2015 – International population reports.
NIH. Issued March 2016. <https://www.census.gov/content/dam/Census/library/publications/2016/demo/p95-16-1.pdf>

Geriatric population worldwide as of 2021

Population aged 65 and older, per cent, 2021

Filter

- aged 0-14
- aged 10-19
- aged 10-24
- aged 15-64
- aged 65+



United Nations Population Fund, accessed January 2022, <https://www.unfpa.org/data/world-population-dashboard>

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General Principles that Affect the Geriatric Patient

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- Effects of Aging
- Multiple co-morbidities
- Poly-pharmacy
- Disability/impaired recovery

In the next few slides we will cover the general principles that affect the geriatric population that impact how we, as clinicians, work with this group of patients.

General Principles: Aging

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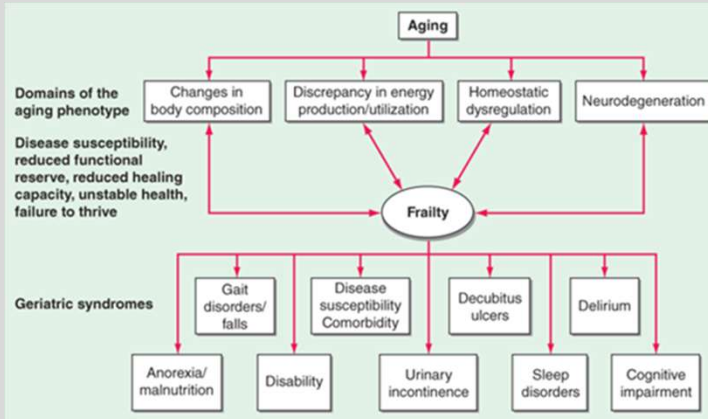
- The inevitable, irreversible decline that occurs in organs over time even in the absence of illness/injury/poor lifestyle choices
- Manifested initially by the reduced capacity of the organ(s) to maintain homeostasis in illness/injury
 - Cardiovascular, renal and CNS systems are especially vulnerable

SOURCE: Greenspan SL, Korytkowski M, Resnick NM. Chapter 23. Geriatric Endocrinology. In: Gardner DG, Shoback D. eds. Greenspan's Basic & Clinical Endocrinology, 9e. New York, NY: McGraw-Hill; 2011.

Model of aging, frailty & geriatric syndromes

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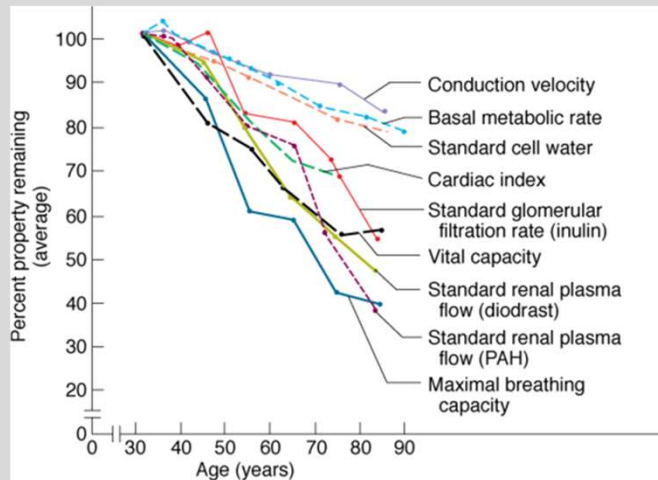


Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. Loscalzo: Harrison's Principles of Internal Medicine, 19th Edition. www.accessmedicine.com
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SOURCE: Ferrucci L, Studenski S. Clinical Problems of Aging. In: Kasper D, Fauci A, Hauser S, Longo D, Jameson J, Loscalzo J. eds. Harrison's Principles of Internal Medicine, 19e. New York, NY: McGraw-Hill; 2015.

Here we see a model that describes how aging leads to changes in body composition, energy production/utilization, homeostasis and neurodegeneration. These in turn lead to frailty which in turn leads to further changes in changes in body composition, energy production/utilization, homeostasis and neurodegeneration AND also contributes to the development of geriatric syndromes which impact the patient's response to disease/illness, as well as healing/recovering from illness.

Effects of aging on physiology



Source: Gardner DG, Shoback D: *Greenspan's Basic & Clinical Endocrinology*, 9th Edition: www.accessmedicine.com
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SOURCE: Greenspan SL, Korytkowski M, Resnick NM. Chapter 23. Geriatric Endocrinology. In: Gardner DG, Shoback D. eds. *Greenspan's Basic & Clinical Endocrinology*, 9e. New York, NY: McGraw-Hill; 2011.

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We can see from this diagram some of the physiologic changes that occur in the aging patient – decrease in nerve conduction velocity, basal metabolic rate, cardiac function, renal function, & respiratory function, etc.

General Principles: Multiple Co-morbidities

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- Geriatric patients often have multiple concomitant illnesses that interact with one another to impact the patient's physiology, presentation of illness, and recovery
- Symptoms of disease often present in the most compromised organ system
 - *Example:* hyperthyroidism in a patient with advanced heart disease could present as atrial fibrillation
 - *Example:* hypothyroidism in a patient with poorly controlled diabetes could present as severe constipation or fecal impaction

SOURCE: Greenspan SL, Korytkowski M, Resnick NM. Chapter 23. Geriatric Endocrinology. In: Gardner DG, Shoback D. eds. Greenspan's Basic & Clinical Endocrinology, 9e. New York, NY: McGraw-Hill; 2011.

General Principles: Poly-pharmacy

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- Aging (and comorbidities) affect a drug's
 - Absorption
 - Volume of distribution
 - Protein binding
 - Elimination
- All of the above lead to fluctuations in therapeutic drug levels & increased risk of under/over-dosing
- In the geriatric patient that takes multiple medications, these effects may mimic or mask signs/symptoms of disease in other systems such as the endocrine system for example

SOURCE: Greenspan SL, Korytkowski M, Resnick NM. Chapter 23. Geriatric Endocrinology. In: Gardner DG, Shoback D. eds. Greenspan's Basic & Clinical Endocrinology, 9e. New York, NY: McGraw-Hill; 2011.

General Principles: Disability/impaired recovery

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- Disability in the geriatric patient is almost always multifactorial:
 - lower body fat composition
 - reduced lean body mass with reduced muscle strength & fitness
 - reduced energy efficiency
 - poor nutritional intake
 - neurologic dysfunction
 - cognitive impairment
- Disability occurs late in the process after all the reserves and ability to compensate have been exhausted
- Disability affects the person's ability to recover and to compensate for changes that have occurred with the illness/injury

Ferrucci L, Studenski S. Clinical Problems of Aging. In: Kasper D, Fauci A, Hauser S, Longo D, Jameson J, Loscalzo J. eds. *Harrison's Principles of Internal Medicine*, 19e. New York, NY: McGraw-Hill; 2015

Aging's physiologic changes to endocrine system

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Menopause, ↓ estrogen and <u>progesterone</u> secretion	↑ Incidence of diabetes
↓ Testosterone secretion	↓ Muscle mass
↓ Growth hormone secretion	↓ Bone mass
↓ Vitamin D absorption and activation	↑ Fracture risk
↑ Incidence of thyroid abnormalities	↑ Fracture risk
↑ Insulin resistance and glucose intolerance	Vaginal dryness, dyspareunia
↑ Bone mineral loss	Changes in skin
↑ Secretion of ADH in response to osmolar stimuli	Tendency toward water intoxication

SOURCE: Greenspan SL, Korytkowski M, Resnick NM. Chapter 23. Geriatric Endocrinology. In: Gardner DG, Shoback D. eds. Greenspan's Basic & Clinical Endocrinology, 9e. New York, NY: McGraw-Hill; 2011.

Here is an example of how the changes that occur with aging can affect the endocrine system.

Other physical/physiologic changes that occur

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- Cardiovascular changes
- Temperature regulation
- Visual Acuity
- Hearing
- Oral changes
- Respiratory changes

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

1. Systolic blood pressure rises as we age, aorta and large arteries stiffen and become atherosclerotic, and a decline in cardiac pacemaker cells affects maximal heart rate and the response to exercise and stress. Cardiac output is decreased and valvular leaflet changes/stiffening can occur. These changes can lead to orthostatic hypotension or to the development of arrhythmias, cardiac murmurs like aortic stenosis or mitral regurgitation. Changes to ventricular compliance and reductions in ventricular filling can lead to a 4th heart sound (which is normal in the older patient).
2. Temperature regulation changes so that the aging patient is more likely to have hypothermia OR may not mount a fever at all during illness.
3. Visual acuity decreases due to the development of cataracts, glaucoma or macular degeneration, OR from the effect of illnesses like diabetes or hypertension, or from the development of presbyopia. These changes affect quality of life and also make the patient increasingly susceptible to falls and injuries.
4. Hearing decreases usually after age 5 yo which can affect communication and quality of life.
5. There is decreased saliva production which affects the oral health of the patient leading to increased cavities, gum disease, etc. Teeth may wear down, fall out or become abraded over time – loss of dental health affects feeding, communication, mental health, and appearance.
6. Lose lung capacity – the chest wall becomes stiffer and harder to move, diaphragmatic

strength decreases, respiratory muscles weaken, lungs lose some of their elastic recoil, surface area for gas exchange shrinks and there can be changes to the thoracic spine that lead to increased kyphosis – these lead to a less effective coughing mechanism and also to an increased predisposition to atelectasis and pneumonia.

Other physical/physiologic changes that occur

- Breasts
- Abdomen/GI
- Genitourinary system
- MSK
- Nervous system

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

1. Breasts in particular in women become smaller, more flaccid and pendulous, and the glandular tissue atrophies leading to ducts that become more palpable as stringy strands. Males may develop gynecomastia or increased breast fullness due to obesity or hormonal changes.
2. Abdominal muscles tend to weaken, fat may accumulate in the lower abdomen and the hips so that the appearance of the patient changes. Aging can blunt symptoms of GI disease – pain may be less severe, guarding and rebound tenderness may be diminished or absent, fever is less pronounced, etc.
3. Sexual interest appears to remain intact but frequency of intercourse decreases. Pubic hair decreases and becomes gray. In males, decreased testosterone levels lead to decrease in penis size, testicles drop lower in the scrotum and erections decrease. The incidence of prostatic hypertrophy and prostate cancer increased. In women, menses ends, and menopause leads to a variety of symptoms including vaginal dryness. The labia and clitoris become smaller, the vagina shortens and the vaginal mucosa atrophies and becomes thinner – these changes lead to increased pain with intercourse. The suspensory ligaments of the adnexa, uterus and bladder may also relax leading to urinary incontinence. Additional things that can contribute to urinary incontinence are decreased innervation and contractility of the bladder muscle, loss of bladder capacity,

urinary flow rate and the ability to inhibit voiding.

4. In the MSK system, there is loss of bone with increased development of osteoporosis and increased risk for fractures as well as a loss of height & the development of kyphosis. There is a 30 to 50% decline in muscle mass and ligaments lose some of their tensile strength and we move and react with less speed and agility. Osteoarthritis affects range of motion and can lead to chronic pain. Tremors and other movement disorders can affect mobility and function.
5. Aging and chronic disease states contribute to neurodegeneration which affects memory, strength, reflexes, balance, coordination. This affects quality of life, mood, and recovery from illness as well.

Care of the Geriatric Patient

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- Clinical elements include:
 - Health history – including frequent medication review
 - Physical exam
 - Assessment of functional status
 - Assessment of goals of care and prognosis
 - Access to care – in particular a team-based approach
 - Long-term planning – including palliative care when needed

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

Caring for the geriatric patient necessitates some changes and adaptations to our usual patterns of seeing adult patients. Mostly these adaptations are to accommodate the needs of the patient and also cover topics that we might not be discussing with our younger patients like long term planning, palliative care, and hospice.

Approach to the Geriatric Patient

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- Office setting
- Communication
- Time
- Records

Office setting – the patient's first interaction with you is when they walk into the office, so it's beneficial to the patient to arrange it to meet the needs of the geriatric patient:

Moderate temperatures – not too cool, not too warm

Brighter lighting – since eyesight is diminished

Less clutter – remove obstacles (from hallways, entrances, exam rooms) that can interfere with walking or with canes, wheelchairs, etc.

Quiet exam rooms that are free from distractions

Chairs with higher seating – easier to get out of

Stool with handrail so patient can get on/off the exam table

Communication: ways to facilitate communication with your patient: Have forms with larger font and mail or email them earlier so that they can complete them without feeling rushed or stressed. Ask them to bring family or friends with them to the appointments to ensure that the care plans are understood, and action is taken on any orders. Have someone available to help the patient with technology that might be used in the office – like sign in kiosks, patient portals, etc. Ask the patient to turn on hearing aids, put on glasses and insert dentures. Face the patient directly and sit at their eye level. Have a clear HIPAA document that covers who can and cannot receive information on the patient – patients in this group often have specific people that they want to be involved but sometimes they assume that you'll just communicate with all their children or family – knowing who is the point person or who you can communicate with or who can assist you and the patient is very important. Make sure that the patient and family/caregiver always have a clear list of current medications with dose, route and frequency and why the medication is

being taken. Additionally, make sure that anything you give the patient that is educational is in a large enough print or in their preferred language.

Time: make sure that appointments have enough time to allow you to do your tasks and allow the patient/family/caregiver time to ask questions and for telling their stories. Adjust your pace so that you're not speaking too rapidly or making the patient feel rushed. If needed, consider scheduling two appointments to allow time for everything that needs to be covered and so it won't also be fatiguing for the patient.

Records: the geriatric patient often has a variety of specialists that are involved in their care, getting these consultation reports is sometimes challenging but keeping records up to date as much as possible means that you have a comprehensive picture of what is happening when the patient is outside of your care. Don't forget to document conversations with specialists, vendors who provide special services like care product deliveries, insurance companies, and with the patient/family and caregivers. Have up to date files on advance directives like DNR/DNI, MOLST forms, healthcare proxy, living wills, etc. will ensure that we comply

with our patients wishes and care instructions.

Approach to the Geriatric Patient: History

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- History taking
 - Use as many sources of information as possible – this prevents repetition of material, reinforces what the patient is stating and provides a more accurate picture of the patient's current status and needs
 - Use direct questions and listen carefully
 - Keep in mind the patient experience and perspective
 - Listen
 - Cultural dimensions –
 - Review Table 27-2 in Bates, 13th edition, chapter 27 for examples of questions related to cultural dimensions

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

Source that are available to us: Family members
Caregivers
Home health aides
Other physicians
Hospital discharge records
Previous medical records

The patient is the one who is experiencing illness/disease states or decline, so please keep that in mind. Ask patients a thorough history and review of systems when it is necessary using direct questions. Sometimes patients may underreport symptoms for fear of losing independence or fear of being diagnosed with an illness that they fear like malignancy; sometimes they have common symptoms or signs but think they are part of aging (like joint pain, back pain, memory deficits, etc.) but that are masking another underlying diseases that is readily treatable. Patients may have cognitive impairment and may not provide an accurate timeline or description of what they are feeling/experiencing because they have forgotten – family, caregivers, etc. can help with this.

Additionally, many geriatric patients' perspective of their life shifts to how many more years do I have left and what will those years look like for me, how can I stay independent and live my own home, who can help me, etc. While it's important to have the family and

caregivers involved, we are still responsible to the patient for their wishes so it's important that they are involved in the decision making process (barring of course something that diminishes their capacity to make decisions).

To avoid making assumptions or missing key points about how the patient views themselves, their condition or their prognosis, make sure to listen to what they are expressing and explore their comments when you note that there is a lack of clarity or something they say may sound confusing. For example, a patient with chronic pain who says "I don't want to live with this anymore" could be expressing that the pain is unbearable OR they could be expressing a depression with suicidal ideation – explore this further to ensure that you both have a shared understanding of what the patient is telling you.

Finally, there is diversity in our aging population in the United States and in other countries around the world reflecting the demographic shifts that have happened with immigration, wars, and conflicts but also with socioeconomic factors like poverty, discrimination, climate changes, education, etc. Culture shapes our beliefs about aging, illness, death, diseases, mental health, choices in doctors and healers, and about the healthcare system in general. It is important to try to understand your patient's cultural norms and to adjust care plans to reflect those whenever necessary. Let patients establish their cultural identity with you – explore where they are from, what are their culture's explanations about illness, dying, aging; what are factors that impact level of function and psychosocial environment and what is the expectation of the clinician patient relationship. This is especially important in discussions about end of life care and decision making.

Approach to the Geriatric Patient: History & physical

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- Special areas of concern
 - Areas that may require additional evaluation or exploration include:
 - Functional impairments in daily living or instrumental activities of daily living
 - Medication management and side effects
 - Smoking
 - Alcohol
 - Nutrition
 - Pain
 - Cognitive impairment
 - Urinary incontinence
 - Falls

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

Approach to the Geriatric Patient: History & physical

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- Activities of daily living
- Instrumental activities of daily living

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

ADLs – ask how well the patient performs these 6 basic self care abilities: bathing, dressing, toileting, transferring, continence and feeding.

IADLs – ask how the patient performs these 8 functions: using a telephone, shopping, preparing food, housekeeping, laundry, transportation, taking medication and managing money or finances.

It's important to know whether the person is independent and can manage on their own without any difficulty OR if someone needs to assist them for whatever reason. These not only give you a sense of the person's ability to be independent and how well they are functioning at that level but can also help you to make decisions about any additional services that they may need (like home care, physical therapy, mobility aids, social services like meals on wheels or transportation to and from medical visits) or if someone needs to step in to assist when the patient can no longer manage or perform these actions on their own.

Approach to the Geriatric Patient: History

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Beers Criteria for Potentially Inappropriate Medication Use in Older Adults

- A list of about 100 medications or medication classes that is published by the American Geriatrics Society approximately every 3 years
- Acts as a guide for healthcare providers to make safe prescribing decisions for geriatric patients
- Includes 5 sections:
 - Medications to avoid for those 65 and older and who are not in hospice or palliative care settings
 - Medications to avoid in people with certain health conditions
 - Medications to avoid that cause drug interactions that are potentially harmful
 - Medications with potential side effects whose risk outweigh the benefits
 - Medications to use in limited doses or avoided due to their effects on kidney function
- 2023 publication: <https://agsjournals.onlinelibrary.wiley.com/doi/epdf/10.1111/jgs.18372>

A caveat to the Beers Criteria: it is not specific to one person; it attempts to provide guidance for a broad group of people. Because of this, it doesn't take into consideration a specific patient's overall health, all underlying medical conditions/situations that may cause a provider to select a specific medication. Therefore, the Beers Criteria is considered a tool that providers can use, but it does not make the final decisions on behalf of a provider. Documenting why a specific medication was chosen will allow others to understand how the medication was chosen and why. Revisiting the continued use of the medication at each visit, including asking about side effects and efficacy, will help to determine if it should be continued or if it should be stopped or changed to something different.

Approach to the Geriatric Patient: History & physical

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- Use validated tools and screening tests such as
 - 10 Minute Geriatric Screener
 - Beck Depression Inventory
 - PHQ2 and PHQ9 for depression
 - GAD-7 for anxiety
 - Alzheimer's disease assessment scale - assess the severity of cognitive and non-cognitive dysfunction
 - Mini Mental State evaluation (MMSE) or the Montreal Cognitive Assessment (MoCA) for screening for memory disorders
 - Timed getup and go test for Fall risk assessment

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

Screening tools: These tools have been tested and proven to be useful for these purposes – you must use the test as it is described, in some cases some minor training is required to ensure the evaluation is reliable. Use of validated screening tools helps us gather information in an organized fashion and helps us make clinical decisions as well as track the progression of illness/disease. Using these ADLs and IADLs along with these screening tools to establish functional status provides a baseline for the patient and allows us to making interventions, identify geriatric syndromes that may be present and helps us to optimize the patient's care and function for as long as possible.

10- Minute Geriatric screener

Problem	Screening Measure	Positive Screen
Vision	Two parts: Ask: "Do you have difficulty driving, or watching television, or reading, or doing any of your daily activities because of your eyesight?" If yes, then: Test each eye with Snellen chart while the patient wears corrective lenses (if applicable).	Yes to question and inability to read >20/40 on Snellen chart
Hearing	Use audioscope set at 40 dB Test hearing using 1,000 and 2,000 Hz.	Inability to hear 1,000 or 2,000 Hz in both ears or either of these frequencies in one ear
Leg mobility-Timed Get Up and Go (TUG) test	Time the patient after asking: "Rise from the chair. Walk 10 feet briskly, turn, walk back to the chair, and sit down."	Unable to complete task in 10 seconds
Urinary incontinence	Two parts: Ask: "In the last year, have you ever lost your urine and gotten wet?" If yes, then ask: "Have you lost urine on at least 6 separate dates?"	Yes to both questions
Nutrition/weight loss	Two parts: Ask: "Have you lost 10 lb over the past 6 months without trying to do so?" Weigh the patient.	Yes to the question or weight <100 lb
Memory	Three-item recall	Unable to remember all three items after 1 minute
Depression	Ask: "Do you often feel sad or depressed?"	Yes to the question
Physical disability	Six questions: Are you able to ... : "Do strenuous activities like fast walking or bicycling?" "Do heavy work around the house like washing windows, walls, or floors?" "Go shopping for groceries or clothes?" "Get to places out of walking distance?" "Bathe, either a sponge bath, tub bath, or shower?" "Dress, like putting on a shirt, buttoning and zipping, or putting on shoes?"	No to any of the questions

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Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021.
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The 10 minute geriatric screener covers 3 areas: cognitive, psychosocial and physical function.

Approach to the Geriatric Patient: History

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Mini Mental Status Exam (MMSE)

- Created in 1975 and validated multiple times in English and other languages
- 11-question measure that tests five areas of cognitive function:
 - orientation, registration, attention and calculation, recall, and language.
- Takes approximately 5 to 10 minutes to administer
- Max score is 30. A score of 23 or lower is indicative of cognitive impairment.
 - Helps to differentiate patients with cognitive impairment from those without cognitive impairment
 - Limitations: relies heavily on speaking, reading and writing. Therefore, patients who hearing or visually impaired, have low English proficiency, or other communication disorders may perform poorly even if they are cognitively intact. Performance can also be impacted by fatigue, lack of sleep, and substance use/abuse.

Folstein, M., Folstein, S.E., McHugh, P.R. (1975). "Mini-Mental State" a Practical Method for Grading the Cognitive State of Patients for the Clinician. *Journal of Psychiatric Research*, 12(3); 189-198.

The Mini-Mental State Exam

Patient _____ Examiner _____ Date _____

Maximum Score

5 ()
5 ()

Orientation

What is the (year) (season) (date) (day) (month)?
Where are we (state) (country) (town) (hospital) (floor)?

3 ()

Registration

Name 3 objects: 1 second to say each. Then ask the patient
all 3 after you have said them. Give 1 point for each correct answer.
Then repeat them until he/she learns all 3. Count trials and record.
Trials _____

5 ()

Attention and Calculation

Serial 7's. 1 point for each correct answer. Stop after 5 answers.
Alternatively spell "world" backward.

3 ()

Recall

Ask for the 3 objects repeated above. Give 1 point for each correct answer.

2 ()

Language

Name a pencil and watch.

1 ()

Repeat the following "No ifs, ands, or buts"

3 ()

Follow a 3-stage command:

"Take a paper in your hand, fold it in half, and put it on the floor."

1 ()

Read and obey the following: CLOSE YOUR EYES

1 ()

Write a sentence.

1 ()

Copy the design shown.



Total Score

ASSESS level of consciousness along a continuum _____

Alert Drowsy Stupor Coma

"MINI-MENTAL STATE," A PRACTICAL METHOD FOR GRADING THE COGNITIVE STATE OF PATIENTS FOR THE CLINICIAN.
Journal of Psychiatric Research, 12(3): 189-198, 1975. Used by permission.

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Folstein, M., Folstein, S.E., McHugh, P.R. (1975). "Mini-Mental State" a Practical Method for Grading the Cognitive State of Patients for the Clinician. *Journal of Psychiatric Research*, 12(3); 189-198.

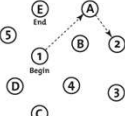
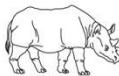
Approach to the Geriatric Patient: History

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Montreal Cognitive Assessment (MoCA)

- A rapid screening instrument for mild cognitive dysfunction.
- Assesses 8 different cognitive domains:
 - attention and concentration, executive functions, memory, language, visuoconstructional skills, conceptual thinking, calculations, and orientation
- Available in approximately 100 languages; training takes about 1 hour
- Validated multiple times for patients age 55 to 85 years old
- Has a variety of versions to meet needs of different patients: telephone only version, blind version, hearing impaired version
- For more information: <https://mocacognition.com/>

MONTREAL COGNITIVE ASSESSMENT (MOCA)						NAME: _____	Date of birth: _____
VISUOSPATIAL / EXECUTIVE						Education: _____	DATE: _____
						Copy cube	Size: _____ (Draw CLOCK (from past eleven) in 3 points)
						10 min	
[] [] [] [] [] [] [] []						/5	
NAMING						Contour	Numbers
  						/3	
[] [] [] [] [] [] [] []						/3	
MEMORY						FACE	VELVET
Read text of names, subject must repeat them. Do both. Do a recall after 5 minutes.						CHURCH	DAISY
[] Total [] Real						RED	No points
ATTENTION						[]	[]
Read text of digits (10 digits / sec.). Subject has to repeat them in the forward order. Subject has to repeat them in the backward order.						[]	[]
Read text of letters. The subject must read with his hand at each letter. A. No points if 2 errors.						[]	[]
Serial 7 subtraction starting at 100						[]	[]
[] 99 [] 86 [] 79 [] 72 [] 65 [] 58						[]	[]
4 or 3 correct subtractions = 3 pts, one correct = 2 pts, correct = 1 pt, correct = 0 pt						[]	[]
LANGUAGE						[]	[]
Repeat : I only know that John is the help today. I ...						[]	[]
The always bill under the couch when dogs were in the room.						[]	[]
Fluency / Name maximum number of names in one minute that begin with the letter f						[]	[]
(N 2 is words)						[]	[]
ABSTRACTION						[]	[]
Similarity between: eg. banana - orange = fruit						[]	[]
[] train - bicycle [] watch - ruler						[]	[]
DELAYED RECALL						[]	[]
Has to read words WITH NO CLUE						[]	[]
FACE VELVET CHURCH DAISY RED						[]	[]
Category rule						[]	[]
Multiple choice rule						[]	[]
Optional						[]	[]
ORIENTATION						[]	[]
[] Date [] Month [] Year [] Day [] Place [] City						[]	[]
Add 1 point if 1/2 or edu						[]	[]
/6						/6	/6
© 2 Montevideo MD - Version November 7, 2004						Normal < 26/30	TOTAL /30
www.mocatest.org						Add 1 point if 1/2 or edu	

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Approach to the Geriatric Patient: History

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Mini-Cog

- A rapid screening instrument for cognitive dysfunction. Not recommended to screen for mild cognitive impairment.
- Test is 3 items: a 3-item recall test for memory and a simply scored clock drawing test:
- Available in multiple languages
- Takes about 3 minutes to administer
- Validated multiple times for patients age 55 to 85 years old
- Has an online version
- For more information: <https://mini-cog.com/step-by-step-mini-cog-instructions/>

Mini-Cog™

Instructions for Administration & Scoring

ID: _____ Date: _____

Step 1: Three Word Registration

Look directly at person and say, "Please listen carefully. I am going to say three words that I want you to repeat back to me now and try to remember. The words are (select a list of words from the versions below). Please say them for me now." If the person is unable to repeat the words after three attempts, move on to Step 2 (clock drawing).

The following and other word lists have been used in one or more clinical studies.^{1,2} For repeated administrations, use of an alternative word list is recommended.

Version 1	Version 2	Version 3	Version 4	Version 5	Version 6
Banana	Leader	Village	River	Captain	Daughter
Sunrise	Season	Kitchen	Nation	Garden	Heaven
Chair	Table	Baby	Finger	Picture	Mountain

Step 2: Clock Drawing

Say: "Next, I want you to draw a clock for me. First, put in all of the numbers where they go." When that is completed, say: "Now, set the hands to 10 past 1."

Use preprinted circle (see next page) for this exercise. Repeat instructions as needed as this is not a memory test. Move to Step 3 if the clock is not complete within three minutes.

Step 3: Three Word Recall

Ask the person to recall the three words you stated in Step 1. Say: "What were the three words I asked you to remember?" Record the word list version number and the person's answers below.

Word List Version: _____ Person's Answers: _____

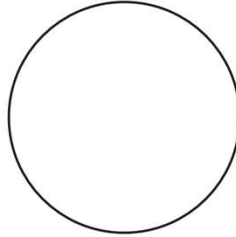
Scoring

Word Recall: _____ (0-3 points)	1 point for each word spontaneously recalled without cueing.
Clock Draw: _____ (0 or 2 points)	Normal clock = 2 points. A normal clock has all numbers placed in the correct sequence and approximately correct position (e.g., 12, 3, 6 and 9 are in anchor positions) with no missing or duplicate numbers. Hands are pointing to the 11 and 1 (10:10). Hand length is not scored. Inability or refusal to draw a clock (abnormal) = 0 points.
Total Score: _____ (0-5 points)	Total score = Word Recall score + Clock Draw score. A cut point of ≥3 on the Mini-Cog™ has been validated for dementia screening, but many individuals with clinically meaningful cognitive impairment will score higher. When greater sensitivity is desired, a cut point of ≥4 is recommended as it may indicate a need for further evaluation of cognitive status.

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Clock Drawing

ID: _____ Date: _____



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Reference: <https://www.alz.org/media/documents/mini-cog.pdf>

Get up and go test

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BOX 27-7

Timed Get Up and Go Test^{86,87}

Performed with patient wearing regular footwear, using usual assistive device if needed, and sitting back in an armless chair. After instructing the patient on what to do, on the word "Go," the patient should:

1. Get up from the armless chair
2. Walk 3 m (in a line)
3. Turn around
4. Walk back to the chair
5. Sit back down

Time the second effort. Observe patient for postural stability, steppage, stride length, and sway.

Scoring:

- Normal: completes task in <10 seconds
- Abnormal: completes task in >20 seconds

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

The 10 minute geriatric screener covers 3 areas: cognitive, psychosocial and physical function.

Approach to the Geriatric Patient: History & physical

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- Physical exam
 - vitals
 - General survey
 - Systems

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

When documenting vitals, pay attention to norms for the patient and any deviation from norms, in particular changes to weight which can be a good indicator that something is wrong or has changed in the patient. When necessary, perform orthostatic measurements of BP and heart rate especially when the patient complains of things like weakness, dizziness, light headedness, syncope, or visual changes with position changes.

General survey: important to pay attention to mental status, appearance (grooming, cleanliness of clothing, shoes), and mobility (how do they get into the room, do they need a mobility aid or do they need assistance).

Perform a physical examination that maintains patient's modesty & respects cultural norms. Remember to remove clothing and evaluate directly on skin to minimize the risk of missing a sign of illness. Provide your patient with a gown and drape and ask the patient if family or caregivers that are with them can stay in the room before you perform the physical exam.

The physical examination generally will entail examining all systems.

Approach to the Geriatric Patient:
SOAP note

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- Use descriptive phrases whenever necessary or the patient's own words if needed
- Share with other members of the healthcare team
 - Include a plan of care for any chronic or acute disease state so that everyone who is involved knows what the goals and the treatment plan are for the patient

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

Approach to the Geriatric Patient: Preventive care/health promotion

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- Screening
- Exercise and other physical activity
- Household safety
- Immunizations
- Elder abuse/mistreatment

Bickley LS, Szilagyi PG, Hoffman RM, Soriano RP. 2021. Bates' Guide to Physical Examination and History Taking, 13e, Chapter 27: Older Adult. Wolters Kluwer, 2021

As more adults live to an older age, in particular those that are 80 yo and older, the decision about when to screen for things like cancer, becomes more complex and requires a discussion about the limitations in this age group, treatment and affects on quality of life if a disease is found. The decision to screen should also include the patient's level of health and function, comorbidities, prognosis, and expectations – age should not be the only factor that is considered. Use screening guidelines where you can to help you have these conversations with your patients.

Other things to screen for are found in the 10 minute geriatric screener and require less complex conversations: vision, hearing, falls, mobility, etc. These should be completed and the patient should be informed of important findings and what the plan is to address them.

Exercise and physical activity are important for maintaining mobility & independence, reducing falls and pain, and keeping the person engaged in activities outside of the home thereby reducing social isolation. Recommendations include aerobic activity and muscle strength training activities for major muscle groups. Obviously these can be adapted depending on the person's ability and if needed a physical therapist should be part of the healthcare team if there are any issues with exercise or physical activity.

House hold safety – often overlooked as we age in our homes, but essential to preventing

injury. Some basic things that can be done in and outside of the home (like the garden or garage): ensure appropriate lighting, get rid of furniture that is too low or too high to get into or out of, install handrails in both directions on staircases, remove clutter & ensure clear paths from one room to another, remove area rugs or tape them down so that they don't move, ensure easy access to telephones to call for help by moving them to low tables or having a mobile phone OR getting the person a clinical alert device that can call for help. Wear well fitting shoes in and out of the house – remove laces and use Velcro if needed to minimize tripping on a shoelace; keep commonly used items within reach to minimize need to stand on a step stool, install grab bars in the tub and shower, use a shower chair if necessary. Remove any cords that could be a tripping hazard and secure all firearms. Finally, have someone or a plan on who will check on the patient on a daily basis or multiple times per day if necessary.

Immunizations that geriatric patients should receive include flu, Dtap, varicella, shingles/zoster, pneumococcal, RSV and Covid-19. It is not enough to ask the patient if they 'are up to date on their vaccines' – you should have a record of when they were administered.

Elder abuse or mistreatment is not always reported because of fear of reprisal or increased abuse – about 90% of the time it is a family member or caregiver who perpetrates the abuse. Although the USPSTF has noted that there is no valid reliable screening tool for abuse it is still important to ask about abuse, neglect, exploitation and abandonment. Ask your patient if anyone has hurt them or abused them in anyway or has restricted access to things like money or food. Pay attention to history and physical exam data that could be indicative of abuse or mistreatment. Self-neglect is also important to pay attention to: this is a patient who threatens their own health/safety by not taking care of themselves or by not following care plans. If there is any suspicion of any of these happening, including self-neglect, the patient should be reported to adult protective services.

Approach to the Geriatric Patient: Preventive care/health promotion

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- **Bone Density**

- Imaging used to screen for osteoporosis and determine fracture risk
 - x ray beams of two different energies (dual x ray absorptiometry or DEXA) are passed through specific bones (hip, lumbar spine, wrist).
 - The bone mineral density (BMD) is calculated for each bone and then converted into values related to the average female/male peak bone mass or to the bone mass relative to the patient's age. These are called T scores and Z scores.
 - T scores and Z scores values are usually based on women though in some places there is sufficient male data; use the same cutoffs for both men and women when male data is not available
- Medicare covers DEXA scans every other year for women 65 yo and older and men 70 yo and older **OR** younger patients if they have certain conditions
- Osteoporosis is more prevalent in women, but men have greater morbidity related to osteoporotic fractures

Reference: <https://www.cdc.gov/nceh/radiation/dexa-scan.html>

The complications of broken bones related to osteoporosis are often severe including increased mortality, increased morbidity such as loss of mobility, pain, loss of independence, increased risk of future fractures, etc.

The World Health Organization (WHO) and the International Society for Clinical Densitometry (ISCD) have defined diagnostic criteria for low bone mass (osteopenia) and osteoporosis based on bone mineral density (BMD) measurements in a dual-energy x-ray absorptiometry (DEXA) scan compared with a reference population that reflects predominantly postmenopausal women.

According to the World Health Organization, osteoporosis is defined based on the following bone density levels:

- A T-score within 1 SD (+1 or -1) of the young adult mean indicates normal bone density.
- A T-score of 1 to 2.5 SD below the young adult mean (-1 to -2.5 SD) indicates low bone mass.
- A T-score of 2.5 SD or more below the young adult mean (more than -2.5 SD) indicates the presence of osteoporosis.

Approach to the Geriatric Patient: Preventive care/health promotion

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- Bone density testing is recommended for
 1. Women 65 yo or older **OR** women who are 50 to 64 with certain risk factors like
 - a) post-menopausal not taking estrogen **OR** who is tall (over 5 ft 7 in) **OR** thin (less than 125 lbs)
 - b) Have a personal or maternal history of hip fracture
 - c) Have a personal history of smoking.
 2. A 70 yo man **OR** man age 50 to 69 yo with risk factors such as clinical conditions associated with bone loss (e.g., rheumatoid arthritis)
 3. Use medications known to cause bone loss, including corticosteroids, anti-seizure medications, certain barbiturates, high-dose thyroid replacement drugs or the birth control injection known as Depo Provera.
 4. Have chronic liver disease, chronic kidney disease or a family history of osteoporosis.
 5. Have certain endocrine disorders such as hyperthyroidism, type 1 diabetes, or hyperparathyroidism, hypogonadism, vitamin D deficiency.
 6. Have experienced a fracture after only mild trauma.
 7. Have x-ray evidence of vertebral fracture or other signs of osteoporosis.

References: <https://www.cdc.gov/nceh/radiation/dexa-scan.html>; <https://www.ncbi.nlm.nih.gov/books/NBK538531/>

Approach to the Geriatric Patient: Preventive care/health promotion

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- **FRAX**
 - The Fracture Risk Assessment Tool
 - Generates a score that is used to determine a person's 10-year risk of major osteoporotic fracture
 - Calculated based on 12 factors:
 - Age (between 40 and 90), Sex, Weight in kilograms (kg), Height in centimeters (cm), Prior fracture, Hip fracture in a parent, Current smoking, Steroid use, Having rheumatoid arthritis, Having a secondary cause of osteoporosis, Drinking three or more alcoholic beverages per day, and T-score at the femoral neck
 - <https://fraxplus.org/calculation-tool>

Reference: <https://www.cdc.gov/nceh/radiation/dexa-scan.html>

The Fracture Risk Assessment Tool was developed in 2008 by a team based within the World Health Organization (WHO) who was collaborating with the Centre for Metabolic Bone Diseases hosted at the University of Sheffield

For patients with osteopenia based on DEXA criteria without prior history of osteoporotic fractures, it is recommended to use the FRAX calculator to determine the decision for treatment.

In patients 50 years and older with osteopenia based on DEXA criteria and a 10-year hip fracture probability 3% or higher or a 10-year major fracture probability 20 % or higher based on the U.S.-adapted WHO absolute fracture risk model warrants treatment.

In a patient with a fragility fracture, a diagnosis of osteoporosis based on DEXA is not needed to initiate treatment.

Summary

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- Geriatrics is the branch of medicine that focuses on the care of the elderly, emphasizing prevention and care of chronic illnesses. The geriatric patient is, by convention, anyone who is ≥ 65 years old.
- Aging is the inevitable, irreversible decline that occurs in organs over time even in the absence of illness, injury, or poor lifestyle choices. This leads to changes in body composition, discrepancy in the production and utilization of energy, homeostatic dysregulation, and neurodegeneration. These changes in turn lead to frailty – increased susceptibility to disease, reduced functional reserves, reduced healing, failure to thrive and a series of geriatric syndromes that include gait disorders/increased falls, anorexia/malnutrition, disability, urinary incontinence, delirium, sleep disorders, and cognitive impairment.
- Geriatric patients often have multiple concomitant illnesses that interact with one another to impact the patient's physiology, presentation of illness, and recovery from illness.
- It's important to know how to work with your patient as they age so that they can remain independent and healthy for as long as they can be.

Questions?

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- If you have any questions, please email me at pryo@nyit.edu

Thank you for listening!

Lecture feedback form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

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